

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Primary engineer for backend, infra, and ML serving on a 3-person team – designed and operated full-stack AI developer tooling end-to-end, including REST/gRPC APIs consumed by external clients and frontend team
- Architected **vLLM inference engine with PagedAttention** for GPU-backed LLM serving – eliminated KV cache fragmentation under concurrent load; continuous batching achieving **zero OOM failures** at production traffic
- Designed and maintained **gRPC inter-service layer** with Python/Protobuf – resolved production deadlock via root-cause tracing and call-sequencing redesign; **100% evaluation success rate at sub-50ms p99**
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch, and circuit breaker – **99.9% uptime, 44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Maintained distributed production systems across Python (Django/FastAPI), PostgreSQL, and ML serving – triaged incidents, documented runbooks, and applied post-mortem learnings; on-call rotation

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Architected event-driven **TypeScript/React system** with observer pattern decoupling UI state propagation – accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**; shipped to stable channel
- Designed **C++ IPC transport layer** with Protocol Buffers – selected Protobuf over custom serialization for schema evolution; shipped to Chrome stable serving **3B+ active users** at **sub-50ms p99, 10K+ req/sec**
- Identified settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** – **96% latency reduction**, zero production regressions
- Collaborated with Chrome infrastructure team on design documents and code reviews at **95% test coverage**; changes adopted into production branch reviewed by senior Chrome engineers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS – selected JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users, 90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC

github.com/pulumi/pulumi

- Submitted Go CLI features and bug fixes for multi-cloud (AWS/Azure/GCP) provisioning to a **24.4K+ star** open-source project – under active review by core maintainers; demonstrates open-source SDK contribution workflow
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Languages: Python, TypeScript, Go, C++, JavaScript, Java, SQL, Bash, Rust

Frameworks & Databases: FastAPI, Django, React, Node.js, REST APIs, gRPC, PostgreSQL, Redis, MongoDB, Pydantic

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, CI/CD, CloudWatch, Prometheus, Grafana

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain, LLM-as-a-judge evaluation

Distributed Systems: Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Event-driven architecture

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor